ORCATHON 2026 · SpaceTech Track

TerraWatch — Satellite Surface Change Detection

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THE PROBLEM

Two satellite passes over the same ground, months apart. A human can say "the forest is gone" — but not at scale, not wi measurement, and not with any distinction between human-driven and natural change.

WHAT IT DOES

One fully-convolutional network (5,639 parameters) takes both frames stacked channel-wise and emits 7 per-pixel logits in a single forward pass. Output: a highlighted change map, the change type, whether a human or nature caused it, and an automated incident report a reviewer could act on.

SIX CHANGE TYPES

Deforestation — human-made	Urban expansion — human-made	Excavation — human-made
New road — human-made	Flooding — natural	Wildfire scar — natural

MEASURED ON 200 HELD-OUT SCENES

generated from a seed range disjoint from the training range — no scene can appear in both.

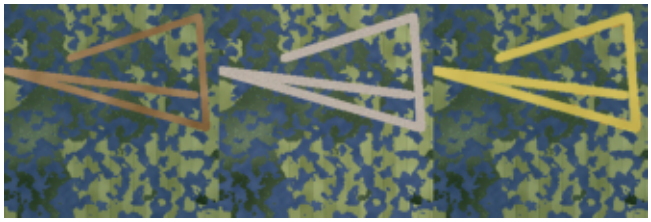
0.905	88.1%	99.3%	5,639
Change IoU	Human vs natural	Pixel accuracy	Parameters
any change vs none			one forward pass

PER-CLASS IoU

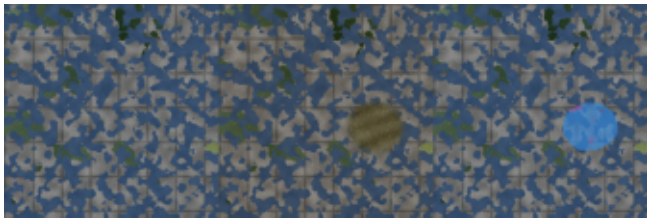
		P	R
New road	**0.944**	0.98	0.96
Wildfire scar	**0.874**	0.93	0.93
Deforestation	**0.753**	0.80	0.92
Flooding	**0.640**	0.85	0.72
Urban expansion	**0.608**	0.81	0.71
Excavation	**0.430**	0.92	0.45

EVIDENCE

Earlier acquisition, later acquisition, and the model's own overlay — from the committed weights.



new road (human-made)



flooding (natural)

github.com/ASylviaofmine/terrawatch · live demo: terrawatch-change-detection.app.workbuddy.host · video: youtu.be/gpFW010gmpE

Built end to end by the author. AI tools assisted with parts of the scripts and documentation; disclosed in the repository README.

Method & the part worth reading

THE DATA IS THE POINT

Training scenes are procedurally generated with exact pixel-level ground truth. That is what makes every number on page 1 a measurement rather than an impression — precision, recall and IoU are all computable because the right answer is known for every pixel.

DEPLOYED WITHOUT PYTORCH

The trained weights are exported to a .npz and the forward pass is re-implemented in NumPy alone, so the serving path has no 2 GB framework underneath it. tools/selftest.py asserts the NumPy output matches the PyTorch model to < 1e-4 — so the hosted demo runs the same model, not an approximation of it.

THE FAILURES THAT MATTERED

The first evaluation returned IoU 0.03 for excavation and 0.00 for flooding, and the obvious move — a bigger network — would have been wrong. The cause was in the data: those textures were identical to the bare soil and the standing water already in the scene, so the image pairs were ambiguous by construction. Fixing the labels took them to 0.57 and 0.64. A later rewrite of the land cover quietly broke urban expansion to 0.07 for the same reason, and was fixed the same way.

WHAT THAT TAUGHT ME

When a class collapses, suspect the labels before suspecting the network. A class is only learnable if its texture cannot be confused with anything else in the scene. This is the habit I carried into the rest of the project — and it is the part I would want to bring to a research group.

PIPELINE



TRAINING DATA

Procedural generator (src/synth.py): five land-cover texture families, six change-event renderers, per-scene illumination and sensor noise, and the ground-truth bookkeeping.

All figures in this sheet come from runs/eval/metrics.json — the same file the repository README is generated from.
github.com/ASylviaofmine/terrawatch